ZEN Univ IUT-1

7th exercises.

1. RHS is even, so $\begin{cases} X & Y & Z \\ \text{odd} & \text{odd} & \text{even} \\ \text{even} & \text{even} & \text{even} \end{cases}$ 2 patterns can be considered. (except symmetricity)

If odd + odd + even, in mod 4,

$$(\text{LHS}) \equiv (\text{odd})^2 + (\text{odd})^2 + (\text{even})^2 \equiv 1 + 1 + 0 \equiv 2$$

$$(\text{RHS}) \equiv 2 \cdot (\text{odd})(\text{odd})(\text{even}) \equiv 0 \quad \#$$

Thus all vars are even.

Let $X = 2x_0$, $Y = 2y_0$, $Z = 2z_0$.

$$2^2(x_0^2 + y_0^2 + z_0^2) = 2^4 x_0 y_0 z_0$$

$$x_0^2 + y_0^2 + z_0^2 = 4x_0 y_0 z_0$$

By same observation, x_0, y_0, z_0 are even.

We can take $x_1 = 2x_0,\ y_1 = 2y_0,\ z_1 = 2z_0$

$$2^2(x_1^2 + y_1^2 + z_1^2) = 4 \cdot 2^3 x_1 y_1 z_1$$

$$x_1^2 + y_1^2 + z_1^2 = 8 x_1 y_1 z_1 .$$

By repeating to apply this, there're positive integers infinite descent,

$$x_0 > x_1 > x_2 > \cdots > 0 \quad \#$$

$\blacksquare$

2. By FLT, $a^{q-1} \equiv 1 \mod q$.

$p-1 \mid q-1$ so for some k, $q-1 = k(p-1)$.

$$a^{q-1} \equiv a^{(p-1)k} \equiv 1^k \equiv 1 \mod p$$

Then $a^{q-1} \equiv 1 \mod pq$ ∎

Quick Review $\gcd(n,m)=1$

$\begin{cases} a \equiv b & \mod n \\ a \equiv b & \mod m \end{cases}$

$\Leftrightarrow \begin{cases} a-b \equiv 0 & \mod n \\ a-b \equiv 0 & \mod m \end{cases}$

$\Leftrightarrow n \mid a-b,\ m \mid a-b$

$\Leftrightarrow nm \mid a-b$

$\Leftrightarrow a \equiv b \mod nm$

3. $p = 2^n + 1$

$\equiv (-1)^n + 1 \mod 3.$

$3 \nmid p$, so $p \equiv 2$ (n must be even).

$\left(\frac{3}{p}\right) = (-1)^{2\cdot(p-1)/4} \cdot \left(\frac{p}{3}\right)$

$= (-1)^{2^{n-1}} \cdot \left(\frac{2}{3}\right)$ by

$n \geq 2$

$= 1 \cdot (-1)$

$= -1$ so 3 is quadratic non-residue in mod p

5.

$$|1001|_3 = 1 \qquad (3 \nmid 1001)$$

$$|2314|_2 = |1157 \cdot 2|_2 = \frac{1}{2}$$

$$\left|\frac{234}{724}\right|_3 = \left|\frac{3^2 \cdot 26}{724}\right| = \frac{1}{9} \qquad (3 \nmid 724)$$

$$\left|\frac{312}{665}\right|_7 = \left|\frac{312}{7 \cdot 95}\right|_7 = 7 \qquad (7 \nmid 312)$$

∎

6. (a) If $a = \frac{n}{m} \in \mathbb{Q} \setminus \mathbb{Z}$, $(\gcd(n,m) = 1,\ m \geq 2)$

m should have at least some prime factor $p \mid m$.

$p \nmid n$, so $|a|_p = \left|\frac{n}{m}\right|_p \geq p > 1$ ※. ∎

(b) $a \in \mathbb{Z}$ by (a)

and $|a|_\infty \leq 1$ means $a = 0$ or 1 or -1 ∎